

Ride Demand & Revenue Optimization Analytics (Uber-like System)

Days of the Week

All

Pickup Location

All

Total Rides

400K

Total Revenue

\$7.30M

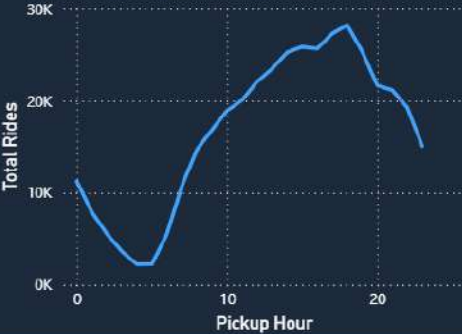
Avg Fare

\$18.26

Revenue per KM

5.34

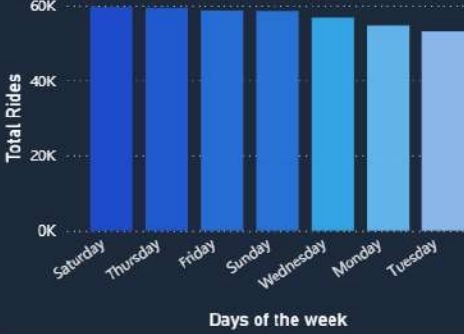
Ride Demand by Hour



Revenue by Hour

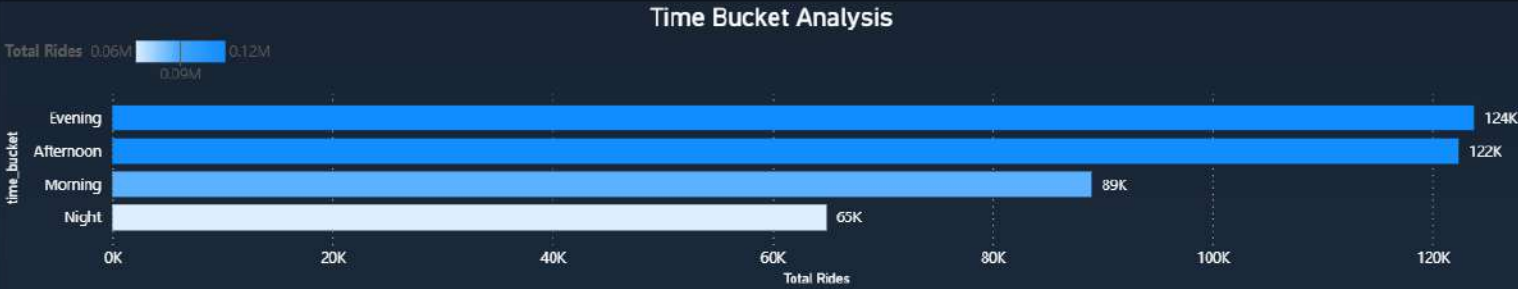


Rides by Day



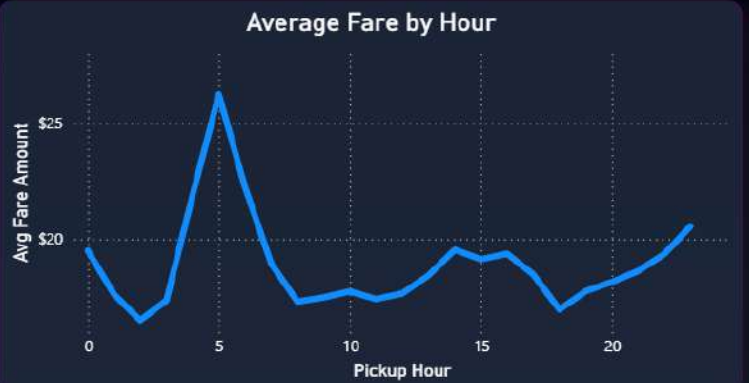
"Ride demand peaks during evening commute hours, while revenue is highest during late-night periods, indicating higher-value trips at night."

Peak Demand Heatmap																									
day_of_week	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	Total
Friday	1450	833	397	242	200	338	812	1723	2144	2430	2654	2780	2928	3162	3638	3627	3561	4133	4097	3930	3213	3265	3620	3293	58470
Monday	1017	473	253	194	189	323	957	1721	2399	2601	2853	2961	3229	3475	3645	3905	3795	3971	3907	3362	2979	2758	2174	1454	54595
Saturday	2798	2289	1688	1086	513	254	394	695	1122	1727	2383	2766	3260	3404	3547	3579	3730	3784	3779	3802	3156	3104	3393	3399	59652
Sunday	3593	3150	2570	1680	884	404	473	749	1022	1663	2517	2877	3436	3611	3745	3827	3852	3472	3341	2919	2492	2459	2092	1515	58343
Thursday	944	486	286	164	176	291	948	2059	2606	2967	2822	2934	3172	3317	3679	3803	3597	4211	4653	3890	3506	3391	3075	2255	59234
Tuesday	612	321	168	131	128	333	953	1923	2682	2769	2893	2919	2999	3158	3423	3507	3391	3759	3929	3389	2945	2948	2252	1451	52983
Wednesday	823	351	167	126	132	302	961	1888	2709	2689	2800	2826	3037	3364	3579	3619	3749	3950	4539	4074	3403	3248	2671	1716	56723
Total	11237	7903	5529	3623	2222	2245	5498	10758	14684	16846	18922	20063	22061	23491	25256	25867	25675	27280	28245	25366	21696	21173	19277	15083	400000



INSIGHT BOX:

Demand peaks during evening hours (5–7 PM), aligning with daily commute patterns and increased ride activity.



Fares rise with distance, with peak pricing in early hours. Most rides are short-distance, as seen in the fare distribution.

Peak Allocation Hour

5 PM

Peak Allocation Score

1.63

Priority Allocation Window

2PM-6PM

Driver Allocation Priority by Hour



Key Insights

- Peak allocation demand occurs between **2 PM–6 PM**
- Friday & Saturday evenings show the highest deployment priority
- Early-morning trips generate higher revenue per ride but insufficient demand for large-scale driver deployment
- Allocation scores above 1 indicate periods requiring additional driver deployment.
Above Line → Increase Drivers and Below Line → Normal Allocation

Recommended Allocation Strategy

- Increase deployment during **2 PM–6 PM**
- Prioritize **Friday & Saturday evenings**
- Reduce excess supply during **2 AM–6 AM**
- Use incentives before peak periods

Demand Pressure by Day & Hour

day_of_week	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Friday	0.66	0.36	0.16	0.10	0.11	0.18	0.42	0.72	0.81	0.97	1.04	1.12	1.18	1.30	1.62	1.61	1.53	1.66	1.53	1.54	1.24	1.33	1.44	1.35
Monday	0.54	0.24	0.11	0.09	0.11	0.18	0.51	0.77	0.99	1.10	1.20	1.18	1.31	1.50	1.66	1.67	1.72	1.70	1.52	1.40	1.28	1.29	1.14	0.81
Saturday	1.11	0.83	0.58	0.39	0.21	0.18	0.23	0.37	0.46	0.63	0.86	1.01	1.22	1.35	1.49	1.49	1.57	1.58	1.52	1.54	1.31	1.31	1.38	1.40
Sunday	1.25	1.08	0.87	0.57	0.36	0.25	0.30	0.39	0.43	0.63	0.94	1.01	1.33	1.48	1.59	1.68	1.68	1.51	1.31	1.24	1.08	1.15	1.07	0.81
Thursday	0.48	0.22	0.12	0.07	0.10	0.17	0.41	0.81	0.94	1.11	1.13	1.19	1.25	1.39	1.66	1.62	1.54	1.68	1.73	1.47	1.39	1.30	1.25	0.99
Tuesday	0.38	0.20	0.10	0.08	0.09	0.21	0.49	0.80	1.13	1.24	1.29	1.32	1.30	1.44	1.63	1.60	1.62	1.69	1.57	1.45	1.27	1.27	1.02	0.81
Wednesday	0.46	0.17	0.07	0.07	0.09	0.17	0.45	0.81	1.09	1.10	1.20	1.15	1.27	1.39	1.57	1.57	1.65	1.64	1.71	1.63	1.39	1.32	1.16	0.85

Peak Surge Hour

5 AM

Peak Surge Multiplier

1.43

Surge Pricing Window

5AM-7AM

Pricing Opportunity by Hour



Key Insights

- Peak pricing opportunity occurs between **5 AM–7 AM**, with the highest estimated surge multiplier observed at **5 AM (1.43x)**
- Early-morning trips generate higher revenue per ride despite lower ride demand.
- Evening periods generate the highest ride volume, while early-morning periods contribute premium fare revenue.

Recommended Pricing Strategy

- Apply moderate surge pricing during **5 AM–7 AM** premium fare windows.
- Increase driver supply during evening peaks instead of relying solely on surge pricing.
- Use dynamic pricing selectively where fare premiums persist despite lower demand.
- Monitor rider response to fare increases and adjust pricing dynamically.

Revenue Opportunity by Day & Hour

day_of_week	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Friday	21.90	19.77	17.07	19.39	34.03	31.69	28.22	18.76	15.27	17.09	16.53	17.36	17.34	18.07	21.22	21.15	19.78	17.24	14.94	16.47	15.79	17.83	16.89	17.91
Monday	27.64	25.01	17.36	23.62	32.80	32.28	28.13	19.68	16.85	17.86	17.69	15.77	16.36	18.42	20.69	18.26	20.31	18.11	15.05	17.18	18.28	21.66	27.21	30.61
Saturday	16.53	13.88	12.43	13.47	17.78	33.49	35.28	30.06	17.65	13.89	13.72	13.99	14.95	16.76	18.65	18.40	18.88	18.50	17.13	17.31	18.30	18.79	17.46	17.95
Sunday	13.83	13.43	13.07	13.09	18.77	43.78	47.43	30.60	20.02	16.32	16.06	14.21	17.15	19.08	20.55	22.02	21.83	21.56	17.59	20.49	21.33	24.81	29.62	32.79
Thursday	27.95	22.89	19.06	22.76	32.06	36.37	20.37	17.07	14.30	15.25	17.46	18.12	16.96	19.14	22.34	19.92	20.21	17.37	15.20	15.61	17.15	16.19	17.99	21.36
Tuesday	34.87	35.12	30.96	29.97	40.33	34.43	23.57	15.32	15.83	17.90	17.89	18.18	16.93	18.64	20.33	18.69	20.37	18.13	14.35	16.32	16.72	16.63	18.52	28.11
Wednesday	33.71	24.11	19.26	33.98	41.07	31.94	22.25	18.50	16.09	16.66	18.20	16.47	17.33	17.00	19.17	18.68	19.28	17.27	14.12	15.88	16.70	16.54	18.97	24.66